WORK AND ENERGY

Class 9 | Science | Chapter 11

WORK

Work is done when a force acts on an object AND the object is displaced in the direction of the force.

W = F x s (Work = Force x Displacement)

SI unit of work: joule (J). 1 joule = work done by 1 N force causing 1 m displacement.

- If force and displacement are in the same direction: positive work.
- If force acts opposite to displacement: negative work (e.g. friction).
- If there is no displacement, or displacement is perpendicular to force: work done = zero (e.g. pushing a wall, carrying a load and walking horizontally).

ENERGY

Forms, Kinetic & Potential Energy

ENERGY

Energy is the capacity to do work. SI unit: joule (J) (same as work).

Common forms: mechanical, heat, chemical, electrical, light, sound, nuclear energy.

KINETIC ENERGY (KE)

Energy possessed by an object due to its motion.

KE = (1/2) x m x v^2

Ex: A moving car, a flowing river, a thrown ball -- all possess kinetic energy.

POTENTIAL ENERGY (PE)

Energy possessed by an object due to its position or configuration/shape.

Gravitational PE = m x g x h

Ex: Water stored in a dam, a stretched bow, a stone held up high.

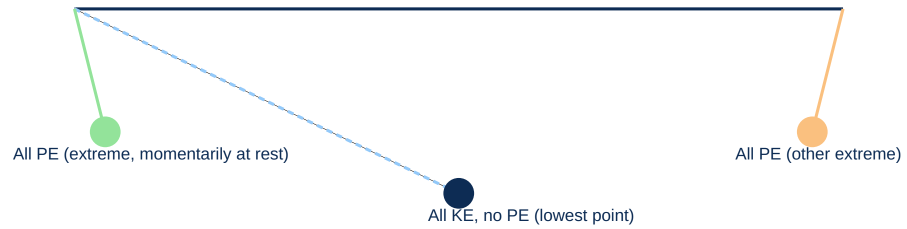
LAW OF CONSERVATION OF ENERGY

Power & Commercial Unit

LAW OF CONSERVATION OF ENERGY

Energy can neither be created nor destroyed; it can only be transformed from one form to another. Total energy before and after transformation remains constant.

DIAGRAM: SIMPLE PENDULUM -- PE <-> KE



Ex: Falling object -- PE converts to KE. Pendulum -- energy oscillates between KE and PE.

POWER

Power is the rate of doing work (or the rate of transfer of energy).

P = W / t

SI unit of power: watt (W). 1 watt = 1 joule/second. Larger unit: kilowatt (kW) = 1000 W.

COMMERCIAL UNIT OF ENERGY

1 kWh (kilowatt-hour, 'unit' of electricity) = 3.6 x 10^6 joules (3.6 million joules).

1 kWh = energy used by a 1000 W appliance running for 1 hour.

SUMMARY & REVISION

Key Formulae + Quick Recap

CHAPTER SUMMARY -- KEY TERMS

- **Work** -> W = F x s, done when force causes displacement
- **Energy** -> capacity to do work
- **Kinetic Energy** -> KE = 1/2 mv^2 (due to motion)
- **Potential Energy** -> PE = mgh (due to position)
- **Power** -> P = W/t, rate of doing work
- **1 kWh** -> 3.6 x 10^6 J -- commercial unit of energy

SOLVED EXAMPLE

Q: A 2 kg object is raised to a height of 5 m (g=10 m/s^2). Find its potential energy.

A: PE = mgh = 2 x 10 x 5 = 100 J.

ONE-LINE RECAP FOR REVISION

Work = Force x Displacement (in the direction of force).

Energy is conserved -- only changes form, total stays constant.

Power = Work/Time; electricity is billed in kWh (1 unit).

Made with love by Examora -- Learn . Practice . Succeed